Exercise: aircraft take-off



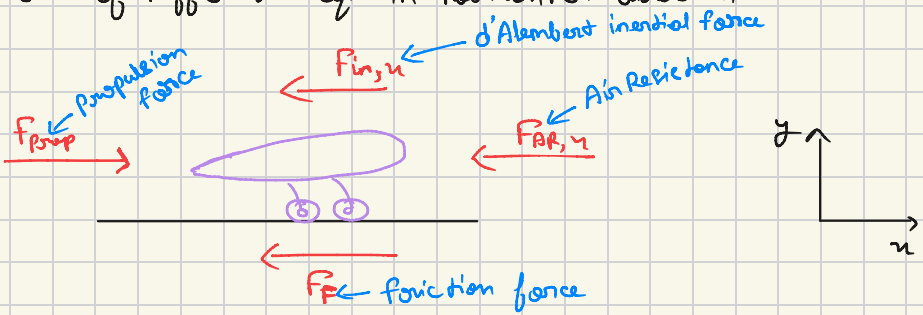
Modelling approach for lift
(depending on the velocity in x-direction)

2-dimensional system
Movement in horizontal and vertical direction
(with coupling between the directions)

$$F_L = \frac{1}{2}\rho_A \cdot c_L \cdot A_{wing} \cdot \dot{x}^2$$

Derive the differential equations of motion for both directions.

Derivation of Differential eqn in horizontal direction



Equilibrium of forces in x-direction:

$$\sum F_x = 0$$

$$F_{prop} - F_{in,x} - F_{AR,x} - F_F = 0$$

(1) Derivation of the differential equation in x-direction

$F_{in,x} = m \cdot a = m\ddot{x}$ (d'Alembert inertial force)

$F_{AR,x} = \frac{1}{2} \cdot \rho_0 \cdot c_d \cdot A_x \cdot v^2 \cdot \sin\vartheta$ (Air Resistance force)

$$= \frac{1}{2} \cdot \rho_A \cdot C_d \cdot A_n \cdot \dot{u}^2 \cdot \sin(\dot{u})$$

$$F_F = \mu \cdot F_N \cdot \text{sign}(v) \qquad \text{(friction force)}$$

from above $F_{in,x}$, $F_{AR,x}$, F_F, we get –

$$\Rightarrow F_{prop} = f_{in,x} + f_{AR,x} + F_F$$

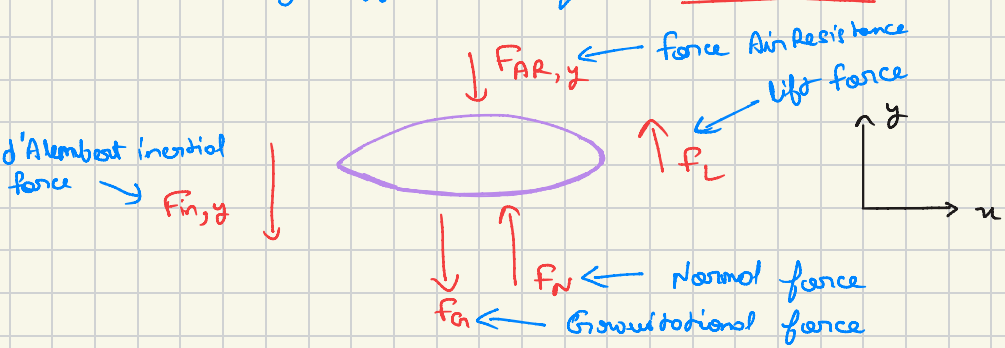
$$\Rightarrow f_{prop} = m\ddot{u} - \frac{1}{2} \rho_A \cdot C_d \cdot A_n \cdot \dot{u}^2 \cdot \sin(\dot{u}) - \mu \cdot F_N \cdot \text{sign}(\dot{u})$$

for Modelling we need explicit form –

$$\ddot{u} = \frac{1}{m} \left[F_{prop} - \frac{1}{2} \cdot \rho_A \cdot C_{d,x} \cdot A_n \cdot \dot{u}^2 \cdot \sin(\dot{u}) - \mu \cdot F_N \cdot \text{sign}(\dot{u}) \right]$$

for Modelling

② Derivation of differential eq. in Y-direction



Equilibrium of forces in vertical direction :

$$\sum F_y = 0$$

$$\Rightarrow \quad F_N + F_L - f_{Gr} - f_{in,y} - f_{AR,y} = 0$$

Derivation of the differential eqn in y-direction

$f_{Gr} = mg$ (Ground additional force)

$f_{in,y} = m \cdot a = m\ddot{y}$ (d'Alembert intential force)

$f_{AR,y} = \frac{1}{2} \rho_A \cdot C_{d,y} \cdot A_y \cdot v^2 \cdot \sin(v)$ (Air Resistance force)

$= \frac{1}{2} \rho_A \cdot C_{d,y} \cdot A_y \cdot \dot{y}^2 \cdot \sin(\dot{y})$

$f_L = \frac{1}{2} \rho_A \cdot C_L \cdot A_x \cdot v^2 = \frac{1}{2} \rho_A \cdot C_L \cdot A_x \cdot \dot{x}^2$ (Lift force)

There are 2-cases of F_N — case ① before takeoff and case ② after takeoff

case ① — Before take-off

$\ddot{y} = 0$ and $\dot{y} = 0$

(acceleration, velocity)

So, $f_{in,y}$ and $F_{AR,y} = 0$

$$F_N + F_L - F_{Gr} = 0$$

$$F_N = F_{Gr} - F_L$$

case ② — condition for take-off

$$F_N = 0$$

Derivation for F_N (normal force) in y-direction

F_N can never be < 0

so, for model

$$F_N = \max\left[(mg - F_L), 0\right]$$

$$F_N = \max\left[\left(m \cdot g - \rho_A \cdot C_L \cdot A_x \cdot \dot{x}^2\right), 0\right]$$

from above F_N, F_L, F_G, $F_{in,y}$, $F_{AR,y}$

$$F_N + F_L - F_G - F_{in,y} - F_{AR,y} = 0$$

$$\max\left[\left(m \cdot g - \frac{1}{2}\rho_A C_L A_x \cdot \dot{x}^2\right), 0\right] + \frac{1}{2}\rho_A \cdot C_P \cdot A_x \cdot \dot{x}^2 - m \cdot g - m \cdot \ddot{y} - \frac{1}{2}\rho_A C_d \cdot A_y \cdot \dot{y}^2 \cdot \sin(\dot{y}) = 0$$

Differential eqn in explicit form for modeling

$$m\ddot{y} = \max\left[\left(m \cdot g - \frac{1}{2}\rho_A C_L A_x \dot{x}^2\right), 0\right] + \frac{1}{2}\rho_A \cdot C_P \cdot A_x \cdot \dot{x}^2 - mg - \frac{1}{2}\rho_A C_{d,y} A_y \dot{y}^2 \cdot \sin(\dot{y})$$

$$\ddot{y} = \frac{1}{m}\left\{\max\left[\left(m \cdot g - \frac{1}{2}\rho_A C_L A_x \dot{x}^2\right), 0\right] + \frac{1}{2}\rho_A \cdot C_P \cdot A_x \cdot \dot{x}^2 - mg - \frac{1}{2}\rho_A C_{d,y} A_y \dot{y}^2 \sin(\dot{y})\right\} - g$$

└ for modeling

From differential eqn in x-direction

$$\ddot{x} = \frac{1}{m} \cdot \left[F_{Prop} - \frac{1}{2} \rho_A \cdot C_{d,x} \cdot A_x \cdot \dot{x}^2 \cdot \sin(\dot{x}) - \mu \cdot F_N \cdot \text{Sign}(\dot{x}) \right]$$

putting the Derivation of the differential eqn of F_N for y-direction.

$$\ddot{x} = \frac{1}{m} \left\{ F_{Prop} - \frac{1}{2} \rho_A \cdot C_{d,x} A_x \dot{x}^2 \sin(\dot{x}) - \mu_r \cdot \max\left[(m \cdot g - F_L), 0 \right] \cdot \sin(\dot{x}) \right\}$$

$$\ddot{x} = \frac{1}{m} \left\{ F_{Prop} - \frac{1}{2} \rho_A C_{d,x} A_x \dot{x}^2 \sin(\dot{x}) - \mu_r \cdot \max\left[\left(m \cdot g - \frac{1}{2} \rho_A C_L A_x \dot{x}^2 \right), 0 \right] \cdot \sin(\dot{x}) \right\}$$